

The empirical recurrence for OEIS A316444, recovered from its tail

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Abstract

OEIS A316444 records “Empirical recurrence of order 69” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 71$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 72$. Its last coefficient is $c_{69} = -15360 \neq 0$, so no further tail satisfies anything shorter; any order-69 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A316444 is “Number of $n \times 4$ 0..1 arrays with every element unequal to 1, 2, 3, 5, 6 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

2, 45, 182, 946, 5256, 26592, 139705, 743357, ...

The entry states:

Empirical recurrence of order 69 (see link above)

As of the “Last modified” line on the live entry (Jul 03 2018 09:10:20, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 71$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 71$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 69: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 69.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 142$ exact terms from $n = 3$ on, it returns order 69 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{69} c_i a(n-i),$$

c_1	5	c_{19}	60946	c_{37}	−555291861	c_{55}	375690235
c_2	5	c_{20}	335234	c_{38}	−808935382	c_{56}	816984803
c_3	13	c_{21}	1140828	c_{39}	−173719079	c_{57}	−368914273
c_4	−146	c_{22}	2674682	c_{40}	749788251	c_{58}	−179878376
c_5	−145	c_{23}	6741594	c_{41}	445013749	c_{59}	85450340
c_6	−35	c_{24}	10578831	c_{42}	7917974	c_{60}	−19220378
c_7	631	c_{25}	1176098	c_{43}	1432058077	c_{61}	9832988
c_8	1884	c_{26}	−16717184	c_{44}	374445297	c_{62}	5226900
c_9	1464	c_{27}	−42775913	c_{45}	−1716121246	c_{63}	−7646868
c_{10}	2856	c_{28}	−39498775	c_{46}	1643841009	c_{64}	2879544
c_{11}	2962	c_{29}	1404287	c_{47}	377428234	c_{65}	−68024
c_{12}	7119	c_{30}	11384703	c_{48}	−4157838406	c_{66}	−685632
c_{13}	−3498	c_{31}	29480232	c_{49}	902319129	c_{67}	248720
c_{14}	−6949	c_{32}	54339218	c_{50}	2354874351	c_{68}	56576
c_{15}	−176550	c_{33}	71505150	c_{51}	−1504172178	c_{69}	−15360
c_{16}	−170918	c_{34}	174309415	c_{52}	675998080		
c_{17}	−348502	c_{35}	131316475	c_{53}	579943520		
c_{18}	128183	c_{36}	−105415027	c_{54}	−1478822230		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 72$, and it is the recurrence OEIS A316444 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{69} - c_1 x^{68} - \dots - c_{69}$. Its constant term is $-c_{69} = 15360$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 69 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 69, so $\deg q = 69$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 69, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 69, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -15360 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-69 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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